



Optimization of therapeutic antibodies by predicting antigen specificity from antibody sequence via deep learning

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The optimization of therapeutic antibodies is time-intensive and resource-demanding, largely because of the low-throughput screening of full-length antibodies (approximately 1×10^3 variants) expressed in mammalian cells, which typically results in few optimized leads. Here we show that optimized antibody variants can be identified by predicting antigen specificity via deep learning from a massively diverse space of antibody sequences. To produce data for training deep neural networks, we deep-sequenced libraries of the therapeutic antibody trastuzumab (about 1×10^4 variants), expressed in a mammalian cell line through site-directed mutagenesis via CRISPR-Cas9-mediated homology-directed repair, and screened the libraries for specificity to human epidermal growth factor receptor 2 (HER2). We then used the trained neural networks to screen a computational library of approximately 1×10^8 trastuzumab variants and predict the HER2-specific subset (approximately 1×10^6 variants), which can then be filtered for viscosity, clearance, solubility and immunogenicity to generate thousands of highly optimized lead candidates. Recombinant expression and experimental testing of 30 randomly selected variants from the unfiltered library showed that all 30 retained specificity for HER2. Deep learning may facilitate antibody engineering and optimization.

The ‘target-to-hit’ stage is a well-established process in antibody drug discovery, as screening hybridomas, phage or yeast display libraries typically results in a number of potential lead candidates. However, the time and costs associated with lead-candidate optimization often take up the majority of the pre-clinical discovery and development cycle¹. This is largely due to the fact that lead optimization of antibody molecules consists of addressing multiple parameters in parallel, including expression levels, viscosity, pharmacokinetics, solubility and immunogenicity^{2,3}. Once a lead candidate has been discovered, additional engineering is often required; phage and yeast display offer a powerful method for high-throughput screening of large mutagenesis libraries ($>1 \times 10^9$); however, they are primarily used only for increasing affinity or specificity to the target antigen⁴. The fact that nearly all therapeutic antibodies require expression in mammalian cells as full-length IgG means that the remaining development and optimization steps must occur in this context. Given that mammalian cells lack the capability to stably replicate plasmids, this last stage of development is done at very low throughput, as elaborate cloning, transfection and purification strategies must be implemented to screen libraries in the maximum range of 1×10^3 , meaning that only minor changes (for example, point mutations) are screened⁵. Interrogating such a small fraction of protein sequence space also implies that addressing one development issue will frequently cause the rise of another or even diminish antigen binding altogether, making multi-parameter optimization very challenging. This challenge frequently results in antibodies with suboptimal biophysical properties for clinical development, which can lead to adverse side effects or even drug failure. For example, self-administered subcutaneous injection of antibodies is becoming an increasingly used approach for patients requiring

frequent dosing, but the identification of highly soluble non-viscous antibodies that retain high biological activity is immensely difficult⁶. The withdrawal of Pfizer’s anti-PCSK9 antibody, bococizumab, from clinical trials is an even more drastic example, where the immunogenicity of the molecule adversely affected long-term treatment efficacy. Conversely, Sanofi and Regeneron’s clinically approved antibody, alirocumab, has the same molecular target of PCSK9 but shows almost no immunogenic effects⁷.

Machine learning applied to biological sequence data offers a powerful approach to augment protein engineering by constructing models capable of making predictions of genotype–phenotype relationships^{8,9}. This is due to the capability of models to extrapolate complex relationships between sequence and function. One of the principal challenges in constructing accurate machine-learning models is the collection of appropriate high-quality training data. Directed-evolution systems are well suited for this as they rely on the linking of biological sequence data (DNA, RNA and protein) to a phenotypic output¹⁰. In fact, it has long been proposed that machine-learning models trained on data generated by mutagenesis libraries be used as a means to guide protein engineering^{11,12}. Gaussian processes, a Bayesian learning model, were recently used to engineer cytochrome enzymes, enabling navigation through a vast protein sequence space to discover highly thermostable variants¹³. Similarly, the design and screening of a structure-guided library of channel rhodopsin membrane proteins was used to train Gaussian process regression models, which were able to accurately predict variants that could express and localize on mammalian cell membranes¹⁴.

In recent years, access to deep sequencing and parallel computing has enabled the construction of deep-learning models capable

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of predicting molecular phenotype from sequence data^{15,16}. For example, deep learning has been used to learn the sequence specificities of RNA- and DNA-binding proteins¹⁷, regulatory grammar of protein expression in yeast¹⁸ and HLA-neoantigen presentation on tumour cells¹⁹. In most cases, deep (artificial) neural networks represent the class of algorithm utilized. Although the complexity of neural networks has changed drastically since their conception, the fundamental concept remains the same: mimicking the connections of biological neurons to learn complex relationships between variables²⁰. As an extension of a single-layer neural network, or perceptron²¹, deep learning incorporates multiple hidden layers to decipher relationships buried in large high-dimensional datasets, such as the millions of reads gathered from a single deep-sequencing experiment. Well-trained models can then be used to make predictions on completely unseen and novel variants. This application of model extrapolation lends itself perfectly to protein engineering because it provides a way to interrogate a much larger sequence space than what is physically possible. For example, even for a short stretch of just ten amino acids, the combinatorial sequence diversity explodes to 1×10^{13} , a size that is nearly impossible to interrogate experimentally.

Here we leverage the power of deep learning to perform multi-parameter optimization of therapeutic antibodies (full-length IgG) directly in mammalian cells (Fig. 1). Starting with a mammalian display cell line²² expressing the therapeutic antibody trastuzumab (Herceptin), we use clustered regularly interspaced short palindromic repeats (CRISPR)–CRISPR associated protein 9 (Cas9)-mediated homology-directed repair (HDR) to introduce site-directed mutagenesis libraries in the variable heavy chain complementarity determining region 3 (CDRH3)²³. To generate information-rich training data, single-site deep mutational scanning (DMS) is first performed²⁴, which is then used to guide the design of combinatorial mutagenesis libraries. An experimental (physical) library size of 5×10^4 variants was then screened for specificity to the antigen human epidermal growth factor receptor 2 (HER2). All binding and non-binding variant sequences were then used to train deep neural networks, which when fully trained and optimized were able to predict antigen specificity based on the antibody sequence with high accuracy and precision. Neural networks were then used to predict antigen specificity on a subset of sequence variants from the DMS-based combinatorial mutagenesis library (approximately 1×10^8 sequences), resulting in $>3.0 \times 10^6$ variants predicted to have a high probability of being antigen-specific. Randomly selected variants were recombinantly expressed and tested, resulting in 30 of 30 showing antigen-specific binding. The *in silico* library of predicted binders were then subjected to several sequence-based *in silico* filtering steps to optimize for developability parameters such as viscosity, clearance, solubility and immunogenicity, resulting in nearly 8,000 antibody sequence variants predicted to have more optimal properties than the starting trastuzumab sequence. Further biophysical characterization of top candidate sequences revealed antibodies with comparable or better properties than trastuzumab for expression and thermal stability, and one candidate that is substantially de-risked for immunogenicity.

Results

DMS determines antigen-specific sequence landscapes and guides rational antibody library design. As the amino-acid sequence of the CDRH3 of an antibody is a key determinant of antigen specificity, we performed DMS on this region to resolve the specificity-determining residues. First, a hybridoma cell line that expressed a trastuzumab variant that could not bind HER2 antigen (mutated CDRH3 sequence) was used (Supplementary Fig. 1). Libraries were generated by CRISPR–Cas9-mediated homology-directed mutagenesis (HDM)²³ by transfecting guide RNA (gRNA) targeting the CDRH3 and a pool of homology templates in the

form of single-stranded oligonucleotides (ssODNs) containing NNK degenerate codons at single sites tiled across CDRH3 (Fig. 2a and Supplementary Fig. 2). The libraries were then screened by fluorescence activated cell sorting (FACS) and populations expressing surface IgG, and which were either antigen binding or non-binding were isolated and subjected to deep sequencing (Supplementary Table 1). Deep-sequencing data were then used to calculate the enrichment scores of the ten positions investigated, which revealed six positions that were sufficiently amenable to a wide range of mutations and an additional three positions that were marginally accepting to defined mutations (Fig. 2b). Although residues 102D, 103G, 104F and 105Y seem to be amino acids of the CDRH3 loop making contact with HER2 (ref. ^{25,26}), 105Y is the only residue that is completely fixed (Fig. 2c). The sequence logo plots generated by DMS were used to guide the rational design of a combinatorial mutagenesis library, which consisted of degenerate codons across all positions (except 105Y; Supplementary Fig. 3 and Supplementary Table 7). Degenerate codons were selected for each position based on their amino-acid frequencies that most closely resembled the degree of enrichment found in the DMS data following one, two and three rounds of antigen-specific enrichment (Supplementary Fig. 2 and equation (2)). This combinatorial library possesses a theoretical protein sequence space of 7.17×10^8 , which is much larger than the single-site DMS library diversity of 191. Libraries containing CDRH3 variants were again generated in hybridoma cells through CRISPR–Cas9-mediated HDM in the same non-binding trastuzumab clone described previously (Fig. 2d). Antigen-binding cells were isolated through two rounds of enrichment by FACS (Fig. 2d and Supplementary Fig. 3) and the binding/non-binding populations were subjected to deep sequencing. The sequencing data identified 11,300 and 27,539 unique binders and non-binders, respectively (Supplementary Table 2). These sequence variants represented only a miniscule 0.0054% of the theoretical protein sequence space of the combinatorial mutagenesis library.

Discriminating between the binding and non-binding sequences in the combinatorial library is challenging at the sequence level. The amino-acid usage per position was comparatively similar between antigen-binding and non-binding populations (Fig. 2e), thus making it difficult to develop any sort of heuristic rules or decipher observable patterns to identify binding sequences.

Training deep neural networks to classify antigen specificity based on the antibody sequence. To learn the high-dimensional patterns that determine antigen binding, we set out to develop and train sequence-based machine- and deep-learning models capable of predicting antibody specificity towards the target antigen HER2. After the deep-sequencing data on binding and non-binding CDRH3 variants were compiled, the amino-acid sequences were converted to an input matrix by one-hot encoding, an approach where each column of the matrix represents a specific residue and each row corresponds to the position in the sequence, thus a ten-amino-acid CDRH3 sequence as here results in a 10×20 matrix. Each row will contain a single '1' in the column corresponding to the residue at that position, and all other columns/rows receive a '0'. We first investigated a wide range of models to assess their accuracy and precision in classifying binders and non-binders from the available sequencing data: one- and two-body logistic regression, *k*-nearest neighbours, support vector machines with linear and Gaussian kernels, standard artificial neural networks, long short-term memory recurrent neural networks (LSTM-RNN) and convolutional neural networks (CNN). The model parameters can be found in Supplementary Table 3; 70% of the original dataset was used to train the models and the remaining 30% was split into a test dataset and validation dataset for model evaluation. Due to the disproportionate amount of non-binding sequences available in the training data, we also investigated the impact of increasing

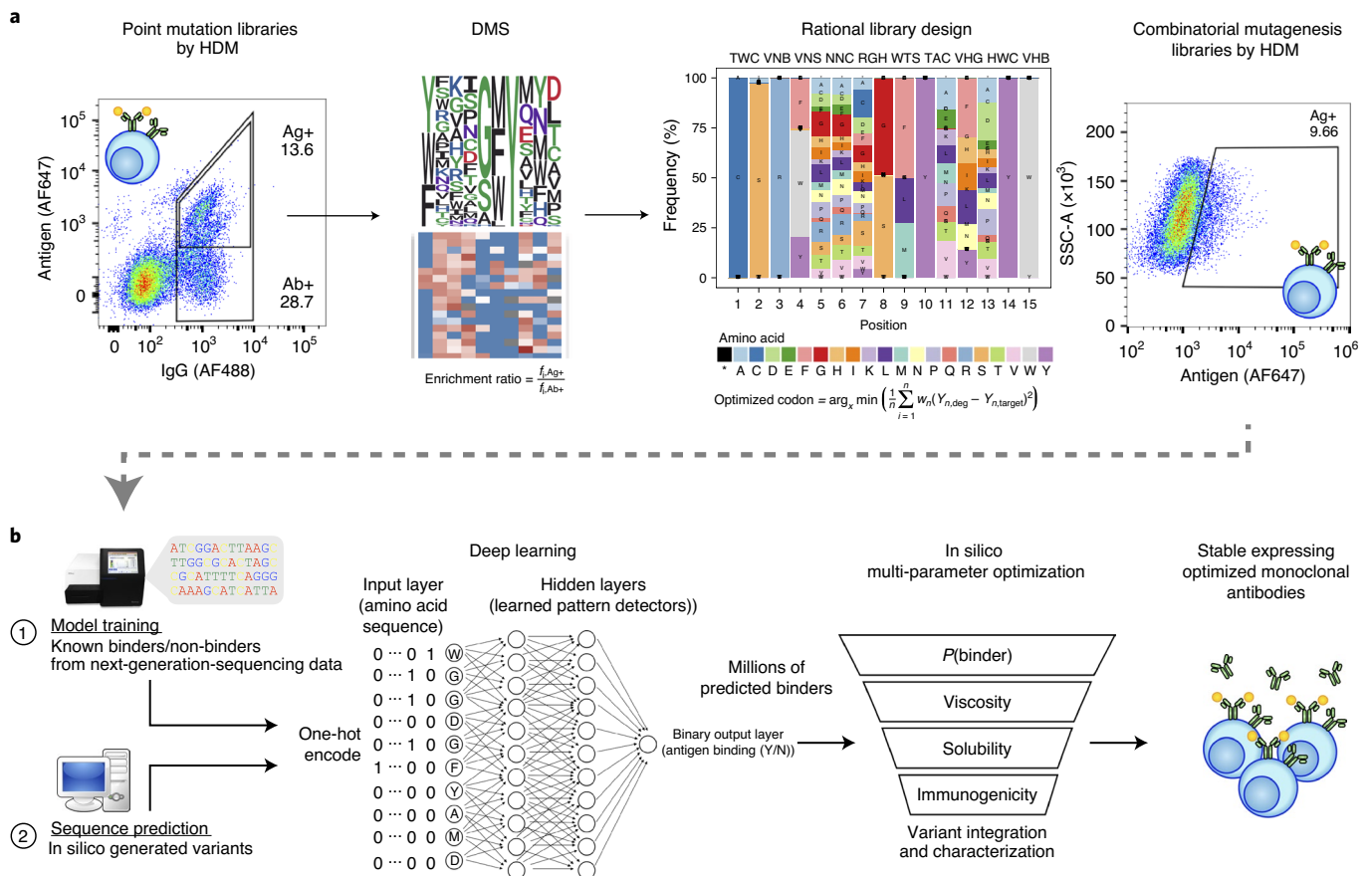


Fig. 1 | Implementing deep learning to predict antibody target specificity. **a**, Generation of quality data capable of training accurate models. First, DMS assesses the impact mutations have on protein function across many different positions. These insights can then be applied to combinatorial mutagenesis strategies to guide protein library design capable of producing thousands of binding variants. The asterisk (*) refers to the stop codon. **b**, Sequence information for binders and non-binders can then be used to train deep neural networks to accurately predict the antigen specificity of unknown antibody variants, producing millions of predicted binders. These binders can then be subjected to any available in silico methods for predicting various developability attributes.

the training data exposed to the models. This subsequently caused an imbalance in the ratio of binding and non-binding sequences exposed to the models, which led to a slight decrease in the model accuracy but an increase in precision (Supplementary Fig. 4). The CNN deep-learning model outperformed the other tested models, so we focused on further optimizing this class of models (Fig. 3a). Convolutional neural networks apply learnable filters to the input data, allowing them to efficiently recognise spatial dependencies associated with a given output. The final model architecture and hyperparameters (Fig. 3b and Supplementary Table 4) were selected by performing a randomized search across various parameters (learning rate, batch size, number of filters, kernel size, stride, activation function, dropout rate and dense layer nodes) using a tenfold cross validation of the dataset (Supplementary Table 4). The final CNN model was trained and evaluated by utilizing all available data points when compiling the training and test datasets (approximate split: 30/70, binders/non-binders). The performance of the CNN was assessed by constructing receiver-operating-characteristic (ROC) and precision-recall curves derived from predictions on the unseen testing datasets. With the proposed training conditions, the CNN was able to accurately classify unseen test data (ROC curve area under curve, 0.91; and average precision, 0.83; Fig. 3c).

A subset of nine sequences from each of the binding and non-binding datasets were expressed and tested for binding against the target antigen via biolayer interferometry (BLI) to further evaluate the false-positive and -negative rates of the model. All of the tested binding sequences retained a markedly high affinity for the

HER2 antigen (Supplementary Fig. 5). However, three of the nine sequences from the non-binding dataset also retained affinity for the HER2 antigen (Supplementary Fig. 6), indicating inaccuracies in the dataset, which could contribute towards potential predictions of false negatives or positives by the model. This could be addressed in the future by employing additional gating strategies during FACS for the generation of the training data.

As a final measure of model validation, neural networks were trained with a dataset containing randomly shuffled binding and non-binding class labels. The model performance of these networks revealed indiscriminate sequence classification on unseen test data (Supplementary Fig. 7), signifying the identification of learned patterns for networks trained with properly classified data.

Predicted binding sequences are recombinantly expressed and antigen-specific. Using our DMS-based combinatorial mutagenesis library as a guide (Fig. 2d), 7.2×10^7 possible sequence variants were generated in silico. The fully trained CNN model was used to classify all 7.2×10^7 sequence variants as either antigen binders or non-binders based on a probability score (P), resulting in a prediction of 10.6×10^6 potential binders ($P > 0.50$). This represented a reasonable fraction (14%) of antigen-specific variants based on experimental screening (Fig. 2d). We increased the prediction threshold for binder classification to $P > 0.7$ to increase confidence. This reduced the antigen-specific sequence space to 6.39×10^6 variants. To validate the precision of our fully trained CNN model, we randomly selected and tested a subset of 30 CDRH3 sequences

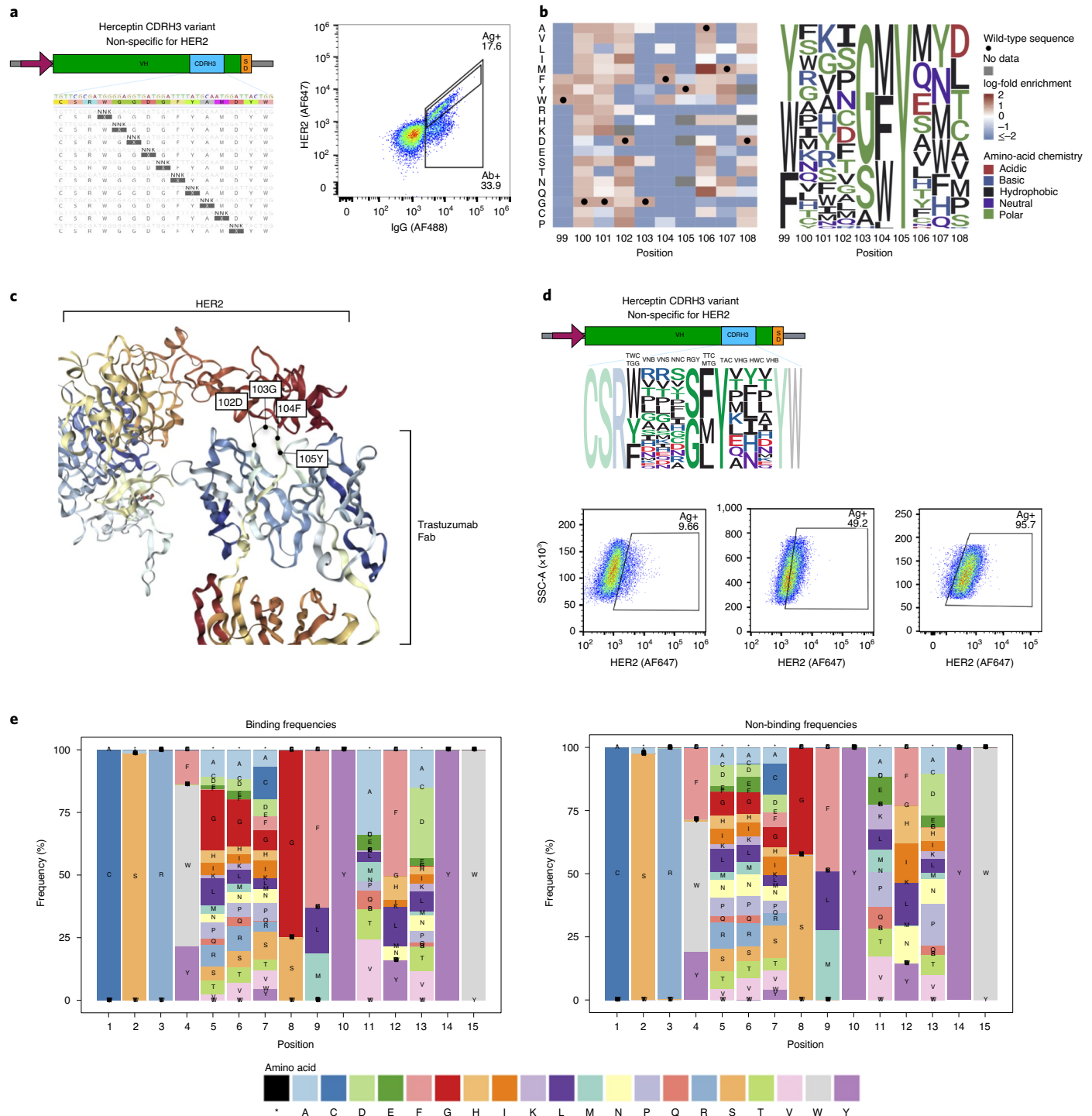


Fig. 2 | Sequence-based analysis of the mutational landscape. **a**, Flow cytometry profile (right) following integration of tiled mutations by HDM (left). Antigen-specific variants underwent three rounds of enrichment (Supplementary Fig. 2). **b**, Corresponding heatmap following sequencing analysis of the pre-sorted (Ab+) and post-sorted (Ag+) populations (left; Supplementary Table 1). Wild-type amino acids are marked by black circles. Resulting sequence logo plot highlighting the positively enriched mutations per position (right). **c**, Three-dimensional protein structure of trastuzumab in complex with its target antigen, HER2 (refs. ^{25,26}). The locations of surface exposed residues 102D, 103G, 104F and 105Y are given. Fab, antigen-binding fragment. **d**, Combinatorial mutagenesis libraries are designed from enrichment ratios observed in the DMS data and integrated into the trastuzumab variant by HDM (top). Amino-acid chemistry legend as per **b**. Flow cytometry plots resulting from transfection of the rationally designed library (bottom). Deep sequencing was performed on the library (Ab+), non-binding variants (Ag−) and binding variants after one and two rounds of enrichment (Ag+1 and Ag+2, respectively; Supplementary Fig. 3 and Supplementary Table 2). **e**, Amino-acid frequency plots of antigen-binding (left) and non-binding (right) variants reveal nearly indistinguishable amino-acid usages across all positions. The asterisk (*) refers to the stop codon.

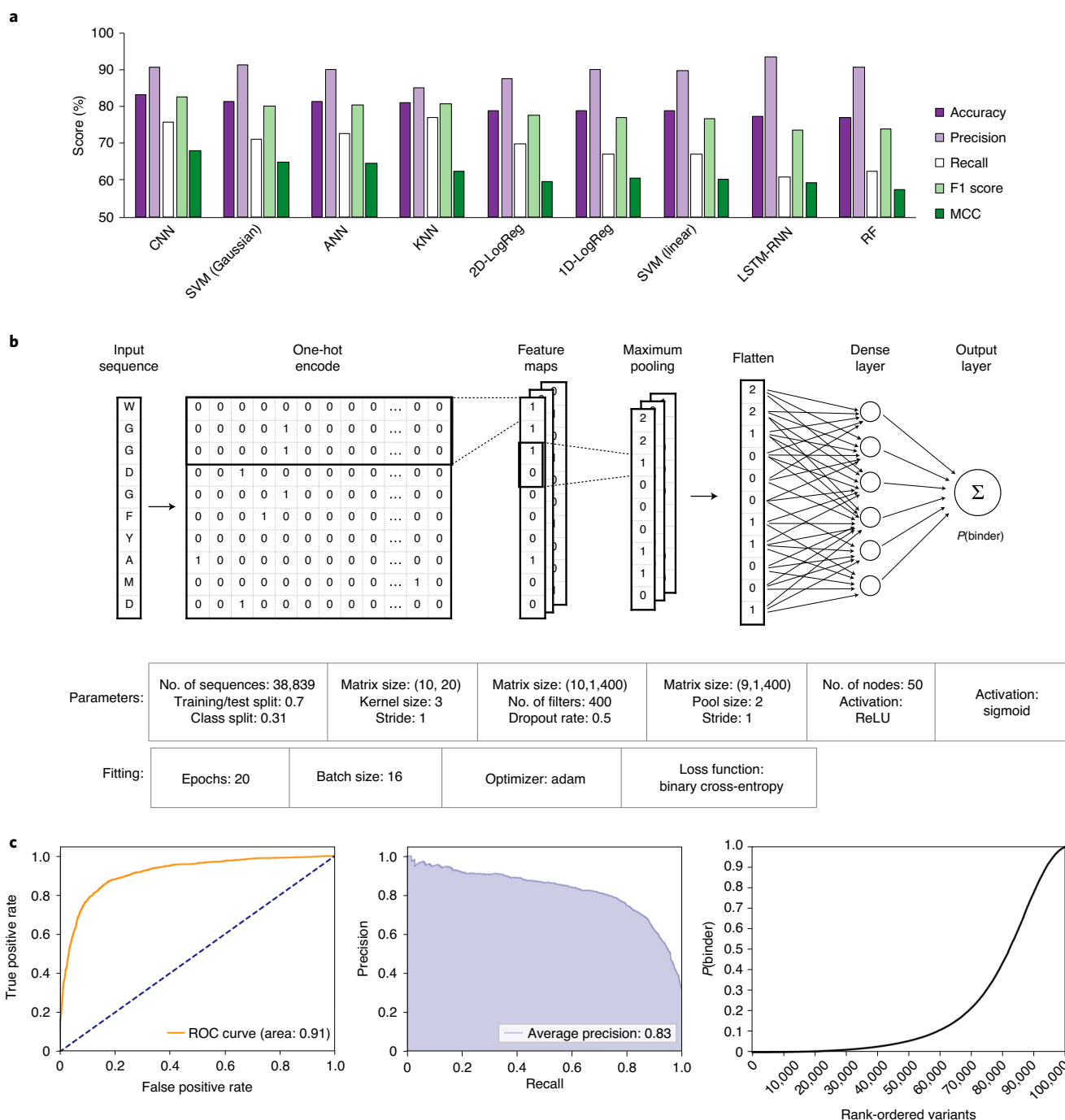


Fig. 3 | Deep-learning models accurately predict antigen specificity. **a**, Model performance metrics for the classification of binding and non-binding sequences across different machine-learning algorithms—that is, CNN, support vector machine (SVM) with a Gaussian kernel, artificial neural network (ANN), *k*-nearest neighbours (KNN), two- and one-body logistic regression (2D-LogReg and 1D-LogReg, respectively), SVM with a linear kernel, long short-term recurrent neural networks (LSTM-RNN) and random forest (RF). The model hyperparameters are given in Supplementary Table 3 and performance metrics (accuracy, precision, recall, F1 score and Matthew correlation coefficient (MCC)) were calculated according to the equations provided in the Methods. **b**, A CNN was selected due to its performance and used for further investigation and optimization. The final architecture and parameters were selected after performing a randomized grid search of the parameters in Supplementary Table 4 using tenfold cross validation. **c**, ROC (left) and precision-recall (middle) curves for the classification of sequences in the test set by the CNN. The high values observed for the area under the curve of the ROC curve and average precision represent robust measures of model accuracy and precision. ROC curve for a random classifier is indicated by the blue dashed line. The distribution of the predicted probabilities ($P(\text{binder})$) for 1×10^5 randomly selected sequences (right).

predicted to be antigen-specific (Fig. 4a) and 12 CDRH3 sequences predicted to be non-binders ($P < 0.1$). To further demonstrate the capacity of deep learning to identify novel sequence variants, we also

added the criteria that the selected variants must have a minimum Levenshtein distance (LD) of five from the original CDRH3 sequence of trastuzumab and contain approximately similar sequences in both

the binding and non-binding training datasets (Fig. 4a,b). CRISPR-Cas9-mediated HDR was used to generate mammalian display cell lines expressing the 42 different sequence variants. Flow cytometry was performed and revealed that all of the 30 predicted binding variants (100%) were antigen-specific (Supplementary Fig. 8) and 11 of the 12 predicted non-binding variants showed no binding activity (Supplementary Fig. 9). Further analysis was performed on the 42 variants to quantify the binding kinetics more precisely via BLI (Fig. 4c and Supplementary Fig. 8,9). The original trastuzumab sequence was measured to have an affinity of 4.0×10^{-10} M (equilibrium dissociation constant, K_d) towards HER2; and although the majority of the binding variants that were tested demonstrated a decrease in affinity, 80% (23/30) were still in the single-digit nanomolar range, 17% (5/30) remained in a potential therapeutic sub-nanomolar range and one variant (3%) showed a near threefold increase in affinity compared with trastuzumab ($K_d = 1.4 \times 10^{-10}$ M; Fig. 4c). Furthermore, 92% of the non-binding variants showed no detectable binding (Supplementary Fig. 9). We also investigated whether the model prediction values and measured affinities were correlated (Supplementary Fig. 10) but no conclusive trends were observed.

Sequence space analysis of deep-learning-predicted variants. We conducted a sequence-similarity network analysis²⁷ of 5,000 randomly selected binding and non-binding sequences to investigate the sequence space of the predicted binding and non-binding variants (Supplementary Table 5 and Supplementary Fig. 11). When similarity networks were generated by clustering CDRH3 sequences with an $LD \leq 3$, we observed 99.7% of all sequences to be within a single cluster, but when the clustering stringency was increased to $LD \leq 2$, the fraction of sequences found within the largest cluster was reduced to 30%, with the majority of other sequences not clustering with any other sequence (Fig. 4d). Although a large portion of the sequences found within the largest cluster were predicted binding sequences, non-binding sequences were also present, illustrating the complexities of the patterns identified by deep neural networks. To further elucidate the high-dimensional patterns of the antigen-binding landscape identified by deep-learning models, we performed the attribution method of Integrated Gradients²⁸ on closely related sequences ($LD \leq 2$; Fig. 4e). This analysis provides a means to visualize nonlinear combinations of amino acids that contribute to classification as a binder or non-binder. This revealed that, unlike position-weight matrices, the CNN model did not equally weight individual residues and positions and thus learned complex nonlinear patterns associated with binding and non-binding.

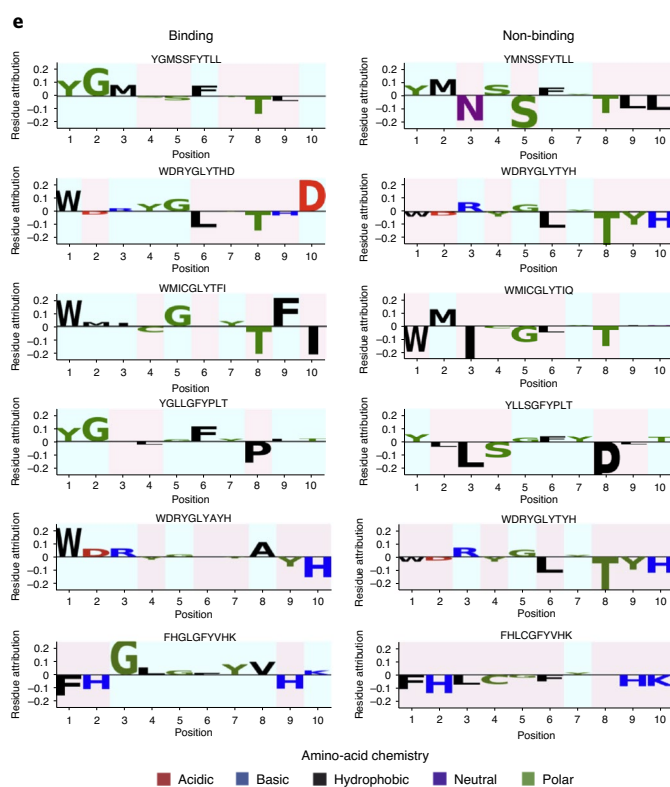
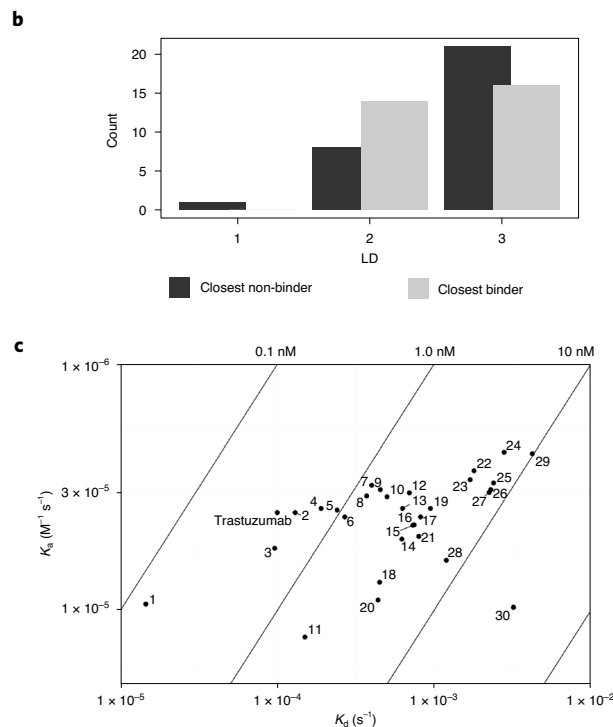
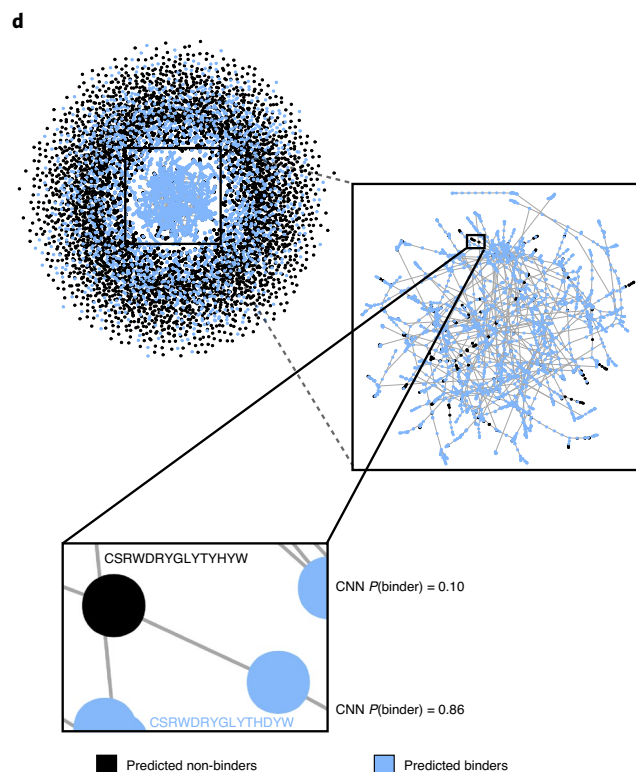
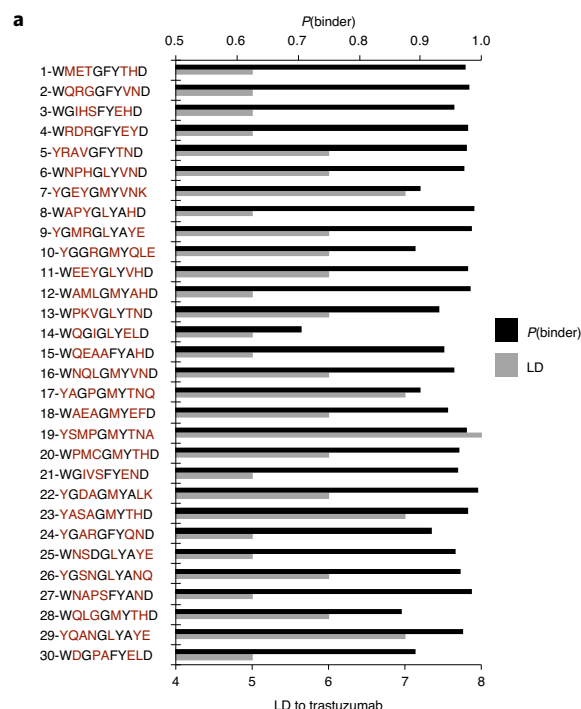
In silico filtering for developability of sequences predicted to bind by deep learning. Next, we characterized all of the 6.96×10^6

deep-learning predicted antigen-specific sequences on a number of parameters to identify highly developable candidates compared with the original trastuzumab sequence. As a preliminary metric, we investigated their sequence similarity to the original trastuzumab sequence by calculating the LD. The majority of sequences showed an edit distance of $LD > 4$ (Fig. 5a). The first step in filtering was to calculate the net charge and hydrophobicity index (HI) to estimate the viscosity and clearance of the molecule². Viscosity was previously shown to decrease with increasing variable fragment (Fv) net charge and increasing Fv charge symmetry parameter (FvCSP); however, the optimal Fv net charge in terms of drug clearance is between 0 and 6.2 with a CDRL1 + CDRL3 + CDRH3 HI sum of less than four. Based on the wide range of values for these parameters in the 6.96×10^6 predicted variants (Fig. 5b,c), we filtered out any sequences that had a FvCSP < 6.61 (trastuzumab FvCSP) or contained an Fv net charge > 6.2 and HI sum that was > 4 or < 0 . This filtering criteria reduced the sequence space substantially to 8.16×10^5 variants. We next padded the CDRH3 sequences with ten amino acids on the amino (N)- and carboxy (C)-terminal ends, and then ran these sequences through CamSol²⁹—a method developed to calculate an overall solubility score based on the physiochemical properties of the amino acids present in a protein sequence and that was shown to correlate well with known developability assays such as ammonium sulfate precipitation and various high-performance-liquid-chromatography methods³⁰. The remaining variants produced a wide range of protein solubility scores (Fig. 5d), and sequences with a score below 0.5 (trastuzumab score) were filtered out, leaving 33,598 candidates for further analysis. As a last step in our in silico screening process, we aimed at reducing immunogenicity by predicting the peptide binding affinity of the variant sequences to MHC class II molecules using NetMHCIIpan³¹. One output from the model is the percentage rank of the predicted affinity of a given peptide compared with a set of 200,000 random natural peptides. Molecules with a percentage rank below two are typically considered strong binders and those with a percentage rank less than ten are considered weak binders to the MHC class II molecules scanned. All possible 15-mers from the padded CDRH3 sequences were run through NetMHCIIpan. After predicting the affinities for a set of 26 HLA alleles determined to cover over 98% of the global population³², sequences were filtered out if any of the 15-mers contained a percentage rank < 5.5 (trastuzumab minimum percentage rank; Fig. 5e). The number of 15-mers with a percentage rank less than ten and the average percentage rank across all 15-mers for the remaining sequences were also calculated. Sequences with more than two 15-mers with a percentage rank < 10 (Fig. 5f) and those with an average percentage rank < 60.56 (trastuzumab average percentage rank) were also filtered out (Fig. 5g).

Fig. 4 | Neural-network-predicted sequences are experimentally validated to be antigen-specific. **a**, To test the precision of the neural network predictions, 30 variants with $LD \geq 5$ to trastuzumab were randomly selected after increasing the prediction threshold ($P(\text{binder}) > 0.75$). Differences to trastuzumab are highlighted in red. These sequences were tested for binding to the target antigen after integrating them into individual hybridoma cell lines by separately transfecting ssODN donor sequences with gRNA. **b**, Levenshtein distance of the closest sequence observed in the binding and non-binding training sets to the 30 randomly selected sequences in **a**. **c**, Affinities for the 30 variant sequences were determined using BLI. Although most sequences display a minor decrease in affinity for the target antigen, the majority of sequences still exude affinities of therapeutic relevance in the single nanomolar (24/30) or sub-nanomolar (5/30) range. K_a , association rate constant; K_d , dissociation rate constant. **d**, A sequence-similarity network analysis was completed on randomly selected variants that were predicted to be binding or non-binding (5,000 in each group) to investigate potential sequence similarities of the classification choices (top). Clustering was performed at an $LD \leq 2$. Similarity network analyses that were performed with additional LDs can be found in Supplementary Table 4 and Supplementary Fig. 12. Although the largest cluster in the network (middle) is composed of 90% predicted binding variants, this comprises only 30% of all sequences in the network. Conversely, 42% of the sequences do not cluster with any other neighbouring sequences, thereby revealing that there are no discernible clusters of binding or non-binding predictions for a large fraction of variants. A zoomed-in view of the largest cluster (bottom) reveals two closely related sequences ($LD = 2$) with opposing binding classifications. **e**, The Integrated Gradients method efficiently extracts and enables visualization of the classification patterns established by the CNN. For this specific example, variants identified in the network with an LD of only 2 were classified as binding (left) or non-binding (right) sequences, respectively. The CNN uniquely identified nonlinear combinations of amino acids that contribute to the sequence's classification as a binder (highlighted with green background shading) or its classification as a non-binder (highlighted with red background shading).

All remaining 7,919 variants contained values equal to or greater than the parameters of the original trastuzumab sequence. When this same filtering scheme was applied to the 11,300 experimentally determined binding sequences (obtained from training/test data), only nine variants remained. Finally, to select and prioritize sequences for experimental characterization that were predicted to have favourable properties across the determined parameters (viscosity, solubility and immunogenicity), we devised and calculated

an overall developability improvement score based on the mean of the normalized values for each relevant parameter (equation (8) in Methods). In this instance, trastuzumab would have a developability improvement score equal to zero. Of the remaining 7,919 predicted binding sequences, 934 variants were identified to have a higher developability score compared with the maximum developability score of the nine experimentally determined binding sequences (Fig. 5h).



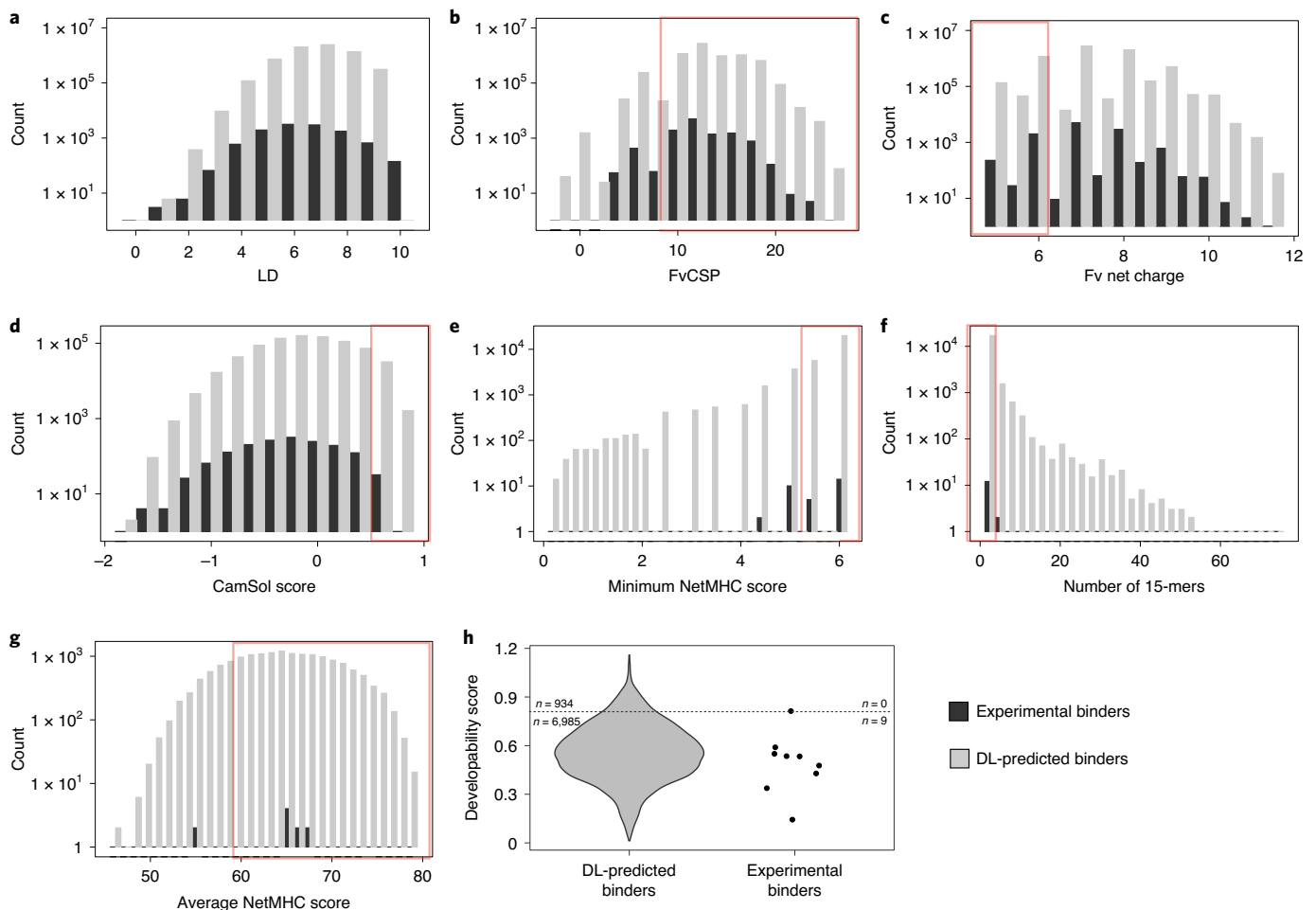


Fig. 5 | In silico screening of the predicted binders identifies candidate sequences for further validation. Antigen-specific variants result in a wide range of in silico-calculated parameters for developability. Staggered histograms show the parameter distributions of all predicted and experimentally observed binders at the different stages of filtering. The red boxes indicate the filtering cut-offs determined by the developability metric calculated for the original trastuzumab sequence. **a**, Levenshtein distance from wild-type trastuzumab. **b**, FvCSP. **c**, Net charge of the Fv domain. **d**, CamSol intrinsic solubility score. **e**, The minimum NetMHCIIpan percentage rank (<2, strong affinity; <10, weak affinity) across all possible 15-mers for a given CDRH3 sequence and across all HLA alleles. **f**, Number of 15-mers found in a given CDRH3 sequence that have a percentage rank of less than ten across all alleles. **g**, Average NetMHCIIpan percentage rank across all possible 15-mers and HLA alleles. **h**, Scatter/violin plot for the overall developability improvement score (equation (3)) of the remaining sequence variants passing all filtering criteria; 293 sequences of the predicted binders have a higher overall developability improvement score than the maximum score identified from an experimental binder indicated by the dashed line. DL, deep learning.

Experimental characterization of computationally optimized candidates identifies high-affinity, high-expressing, thermally stable and de-immunized antibodies. We ordered 100 of the top-ranking CDRH3 candidates according to their developability scores as a single oligo pool and used it to generate a library of stable cell lines with precisely defined sequences by CRISPR-Cas9 HDR. From this library, single-cell sorting was performed and monoclonal variants were further characterized. The affinities of 55 uniquely identified variants (Supplementary Table 6) were similar to those observed in Fig. 4a: 13% sub-nanomolar, 73% single-digit nanomolar and 14% double-digit nanomolar (Fig. 6a). As the CamSol method was previously shown to correlate well with high-performance-liquid-chromatography methods for developability³⁰, we further characterized the top ten variants with the highest affinity for expressibility, thermal stability and immunogenic potential. Expression was measured directly from the cell supernatant via BLI and showed varying degrees of antibody titres, with five variants displaying comparable or better expression than trastuzumab (Fig. 6b). The antibody variants were purified by protein A chromatography and then exposed to thermal ramping with

fluorescence measurements to determine thermal stability (Supplementary Fig. 12). All ten variants again displayed thermal stability that was comparable or better than trastuzumab (Fig. 6c). Finally, 15-mer peptides of select variants were assessed for their potential to elicit T-cell activation via T-cell proliferation assays with donor-matched peripheral blood mononuclear cells. We selected 15-mer peptides for each variant and wild-type trastuzumab according to the regions predicted to have the highest immunogenic potential using NetMHCII (Supplementary Fig. 13). Variants 1 and 3 displayed no significantly responding donors (analysis of variance (ANOVA) $P \leq 0.05$) compared with trastuzumab, which had three significantly responding donors for one 15-mer. Remarkably, variant 1 showed no indications of immunogenic potential across all metrics (number of responders, percentage antigenicity, response index and strength of response; Fig. 6d).

Discussion

To address the limitation of antibody optimization in mammalian cells, we developed an approach based on deep learning that enabled us to identify antigen-specific sequences with high

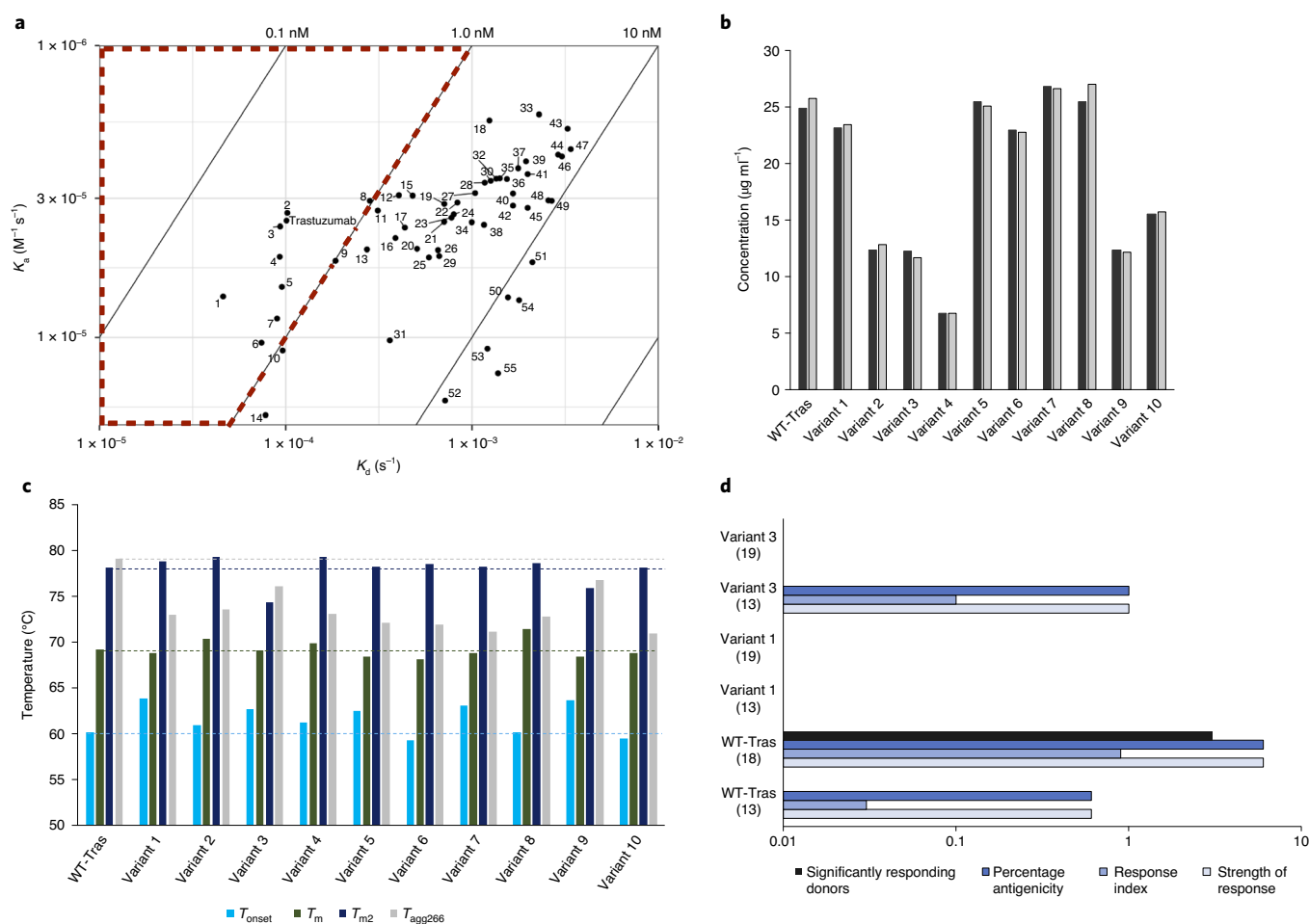


Fig. 6 | Experimental characterization of selected sequences reveals optimal candidates. Monoclonal variants were generated by integrating a pool of the top selected variant sequences identified (Fig. 5h) by HDR, followed by single-cell sorting. **a**, The affinities of 55 unique variants (Supplementary Table 6) were determined using BLI. As observed when validating the models, the majority of variants retained nanomolar affinities and a subset retained sub-nanomolar affinities. The top ten variants with the highest affinity (outlined by the dashed red box) were selected for further characterization. **b**, Antibody expression of the individual variants was also performed using BLI by comparing the initial slope of the binding curve of each variant to a known standard curve ($n=2$). **c**, The thermal stability of the ten variants was assessed through thermal ramp experiments. All variants exuded a similar or better thermal stability (T_{onset} , onset of thermal melting temperature; T_m , thermal melting temperature; T_{m2} , thermal melting temperature 2; T_{agg266} , thermal aggregation temperature) compared with the wild-type trastuzumab antibody (values indicated by dashed lines). **d**, Immunogenic potential of 15-mer peptides from selected variants and wild-type trastuzumab assessed using a T-cell proliferation assay. The starting location of the two 15-mer peptides assessed for each variant (indicated in parentheses) was selected based on regions with high immunogenic potential according to NetMHCII (Supplementary Fig. 13). Variants 1 and 3 show substantially less immunogenic risk according to their potential T-cell epitopes compared with wild-type trastuzumab, which shows antigenicity for both of the tested regions and four (of 20) significantly responding donors for one of the regions. WT-Tras, wild-type trastuzumab.

precision. Calculation and prediction of various biophysical properties of antigen-specific variants allows for the efficient identification of the most developable antibody molecule, resulting in substantial savings in time and cost, and a substantial reduction in the risk for downstream clinical development. Using the clinically approved antibody trastuzumab, we performed single-site DMS, followed by combinatorial mutagenesis to determine the antigen-binding landscape of CDRH3. This DMS-based mutagenesis strategy is crucial for attaining high-quality training data that are enriched with antigen-binding variants, which in this case was nearly 10% of our library (Fig. 2d). In contrast, if a completely randomized combinatorial mutagenesis strategy was employed (that is, NNK degenerate codons), it would be unlikely to produce any substantial fraction of antigen-binding variants. Other approaches to mutagenesis that generate enriched training data³³—such as, shotgun scanning mutagenesis³⁴, binary substitution³⁵ and

recombination^{14,36}—could also be explored in the future for training deep neural networks.

Our initial single-site DMS libraries screened for enriched mutations through antigen binding; however, combining these mutations in a cohesive manner to alter biophysical properties while retaining high antigen affinity is challenging. The amino-acid composition of the binding and non-binding variants is highly similar (Fig. 2e), and visual identification of the sequence patterns that lead to binding is a daunting, if not impossible, task, further exemplifying the value of deep neural networks that are able to learn the high-dimensional space of antigen-binding sequences (Fig. 4e).

A notable finding in this study was that experimental screening of a library of only 5×10^4 variants, which reflected a tiny fraction (0.0054%) of the total sequence diversity of the DMS-based combinatorial mutagenesis library (7.17×10^8), was capable of training accurate neural networks. This suggests that physical library-size

limitations of mammalian expression systems (or other expression systems such as phage and yeast) and deep-sequencing read depth will not serve as a limitation for deep-learning-guided protein engineering.

In the present study, we selected CNNs as the basis of our classification models, as they represent a state-of-the-art approach in deep learning. Other machine-learning approaches—such as *k*-nearest neighbours, random forests and support vector machines—are also well suited for the identification of complex patterns from input data and should also be investigated in any application of protein engineering. But as dataset sizes continue to grow, as is readily achievable with deep sequencing, deep neural networks tend to outperform these classical techniques and are likely to enable the interrogation of more protein sequence space¹⁷. Furthermore, deep generative modelling methods such as variational auto-encoders and generative adversarial networks may also be used to explore the mutagenesis sequence space from directed evolution³⁷.

We generated approximately 7.2×10^7 CDRH3 variants in silico from DMS-based combinatorial diversity and used fully trained CNN models to classify each sequence as a binder or non-binder. The 7.2×10^7 sequence variants comprise only a subset of the potential sequence space and were chosen to minimize the computational effort; however, it still represents a library size that is several orders of magnitude greater than what is experimentally achievable in mammalian cells. We envision that the screening capacity could be easily extended through script optimization and employment of parallel computing on high-performance clusters. The CNN predicted approximately 14% of the classified variants to bind the target antigen, showing exceptional agreement with the frequencies determined experimentally by flow cytometry (Fig. 2d). To experimentally validate the precision of neural networks in predicting antigen specificity, we randomly selected and expressed 30 variants from the library of sequences with a minimum edit distance of five from trastuzumab. The precision of the CNN model was estimated to be approximately 85% (at $P > 0.75$) for each, according to predictions made on the test datasets (Fig. 3b). We experimentally validated that all randomly selected (30/30) antigen-predicted sequences were indeed binders, and several had high affinities. Although we anticipate false positives would be observed by increasing the sample size tested, validation of this subset strongly infers that potentially thousands of optimized lead candidates maintain a binding affinity in the range of therapeutic relevance while also containing substantial sequence variability from the starting trastuzumab sequence. Future work to increase the stringency of selection during screening or a more detailed investigation of correlations between prediction probability and affinity could prove insightful towards retaining high target affinities. Experimental validation of the accuracy of the model to predict the binding status of sequence variants led us to take a more in-depth look at the sequence space of the predicted binding and non-binding variants. A sequence-similarity network analysis at various LDs revealed no distinct clusters between binding and non-binding sequences, indicating an overall sequence similarity of both classifications. By then quantitatively analysing the neural network predictions, we were able to shed light on the high-dimensional patterns captured by the respective models and decipher amino-acid combinations contributing to the classification of a sequence.

Once the affinity of an antibody for its target antigen is within a desirable range for efficacious biological modification, addressing other biophysical properties becomes the focus of antibody development. With recent advances in computational predictions^{38,39}, a number of these properties—including viscosity, clearance, stability², specificity⁴⁰, solubility²⁹ and immunogenicity³¹—can be approximated from the sequence information alone. With the goal of selecting antibodies with improved characteristics, we subjected the library of predicted binders to a number of these in

silico approaches so as to provide a prioritization or ranking structure to select candidates for further biophysical characterization (Fig. 5). After implementing these methods to remove variants with a high likelihood of having poor viscosity, clearance or solubility as well as those with high immunogenic potential, nearly 8,000 multi-parameter optimized antibody variants remained with developability scores greater than the original trastuzumab sequence. Although a limited number of developable sequences can be initially identified experimentally (Fig. 5h), this only reflects a small fraction of the highly developable sequence space (0.2%). By screening in silico libraries, every sequence variant within the defined space is guaranteed, ensuring the presence of multi-parameter optimized sequences. Future work to apply more stringent or additional filters that address other developability parameters (for example, stability, specificity and humanization) could also be implemented to further reduce the sequence space to the most developable therapeutic candidates across even more parameters. For instance, previous studies have investigated the likeness of therapeutic antibodies to the human antibody repertoire⁴¹.

From the library of computationally optimized CDRH3 sequences, we further expressed and characterized 55 unique variants. The variants with the highest affinity were characterized further, from which we identified a sequence (variant 1) that had comparable expression and improved thermal stability relative to trastuzumab. Remarkably, this variant showed substantially de-risked immunogenic potential compared with the original trastuzumab sequence (Fig. 6d). Although trastuzumab does not clinically suffer from immunogenicity-related efficacy problems, it is known that 8–16% of patients develop anti-drug antibodies against it^{42,43}. By having such a diverse sequence space available for functional binding, we are able to recode and remove observable T-cell epitopes from the antibody sequence, while still maintaining functional binding and the desirable biophysical characteristics. Identification of such sequences greatly reduces the risk of anti-drug antibody formation following treatment, leading to prolonged therapeutic efficacy and decreased discontinuation rates. We also envision that this approach could enable the optimization of other functional properties of therapeutic antibodies, such as pH-dependent antibody recycling⁴⁴ or affinity/avidity tuning^{45,46}. This approach can be extended to other regions across the variable light- and heavy-chain genes (for example, the variable light chain complementarity determining region 3; see Supplementary Note). Eventually, multiple regions could be combined, yielding deep neural networks that are able to capture long-range complex relationships between an antibody and its target antigen. To explore these patterns in greater depth, it may be useful to compare neural network predictions with other advanced structural modelling techniques such as those that take advantage of geometric deep learning⁴⁷.

Methods

Mammalian cell culture and transfection. Hybridoma cells were cultured and maintained according to the protocols described previously²³. Hybridoma cells were electroporated with the 4D-Nucleofector System (Lonza) using the SF Cell Line 4D-Nucleofector X kit L or X kit S (Lonza, V4XC-2024 and V4XC-2032) with the programme CQ-104. The cells were prepared as follows: the cells were isolated and centrifuged at 125g for 10 min, washed with Opti-MEM I reduced serum medium (ThermoFisher, 31985-062) and centrifuged again with the same parameters. The cells were resuspended in SF buffer (as per the kit manufacturer's guidelines), after which Alt-R gRNA (IDT) and ssODN donor (IDT) were added. All of the experiments performed utilized constitutive expression of Cas9 from *Streptococcus pyogenes* (SpCas9). Transfections of 1×10^6 or 1×10^7 cells were performed in 100- μ l single Nucleocuvettes with 0.575 or 2.88 nmol Alt-R gRNA and 0.5 or 2.5 nmol ssODN donor, respectively. Transfections of 2×10^5 cells were performed in 16-well, 20- μ l Nucleocuvette strips with 115 pmol Alt-R gRNA and 100 pmol ssODN donor.

Flow cytometry analysis and sorting. Flow cytometry-based analysis and cell isolation were performed using the LSR Fortessa (BD Biosciences) and SH800S (Sony) systems, respectively. When labelling with fluorescently conjugated antigen

or anti-IgG antibodies, the cells were first washed with PBS and then incubated with the labelling antibody and/or antigen for 30 min on ice, protected from light, washed again with PBS and finally analysed or sorted. The labelling reagents and working concentrations are described in Supplementary Table 5. When the cell numbers differed from 1×10^6 cells, the antibody/antigen quantities and incubation volumes were adjusted proportionally. A typical flow cytometry gating strategy for antigen-positive variant selection is shown in Supplementary Fig. 12.

Sample preparation for deep sequencing. Sample preparation for deep sequencing was performed similarly to the antibody-library-generation protocol of the primer extension method described previously⁴⁸. Genomic DNA was extracted from $1-5 \times 10^6$ cells using a Purelink Genomic DNA mini kit (ThermoFisher, K182001). The extracted genomic DNA was subjected to a first PCR step. Amplification was performed using a forward primer binding to the beginning of the VH framework region and a reverse primer specific to the intronic region immediately 3' to the J segment. The PCRs were performed with Q5 high-fidelity DNA polymerase (NEB, M0491L) in parallel reaction volumes of 50 µl with the following cycle conditions: 98 °C for 30 s; 16 cycles of 98 °C for 10 s, 70 °C for 20 s and 72 °C for 30 s; final extension at 72 °C for 1 min and storage at 4 °C. The PCR products were concentrated using DNA clean and concentrator (Zymo, D4013), followed by 0.8× SPRIselect (Beckman Coulter, B22318) left-sided size selection. The total PCR product was amplified in a second PCR (PCR2) step, which added extension-specific full-length Illumina adaptor sequences to the amplicon library. Individual samples were Illumina-indexed by choosing from 20 different index reverse primers. The cycle conditions of PCR2 were as follows: 98 °C for 30 s; two cycles of 98 °C for 10 s, 40 °C for 20 s and 72 °C for 1 min; six cycles of 98 °C for 10 s, 65 °C for 20 s and 72 °C for 1 min; 72 °C for 5 min; and 4 °C storage. The PCR2 products were concentrated again using DNA clean and concentrator and run on a 1% agarose gel. Bands of the appropriate size (approximately 550 bp) were gel-purified using the Zymoclean gel DNA recovery kit (Zymo, D4008). The concentrations of the purified libraries were determined using a Nanodrop 2000c spectrophotometer and pooled at concentrations aimed at optimal read return. The quality of the final sequencing pool was verified on a fragment analyser (Advanced Analytical Technologies) using a DNF-473 standard sensitivity NGS fragment analysis kit. All of the samples that passed the quality control were sequenced. Antibody library pools were sequenced on the Illumina MiSeq platform using the reagent kit v3 (2 × 300 cycles, paired-end) with 10% PhiX control library. The base-call quality of all samples was in the range of a mean Phred score of 34.

Bioinformatics analysis and graphics. The MiXCR v2.0.3 programme was used to perform data pre-processing of raw FASTQ files⁴⁹. The sequences were aligned to a custom germline gene reference database containing the known sequence information of the V- and J-gene regions for the variable heavy chain of the trastuzumab antibody gene. Clonotype formation by CDRH3 and error correction were performed as described previously⁴⁹. Functional clonotypes were discarded if a duplicate CDRH3 amino-acid sequence arose from MiXCR uncorrected PCR errors or a clone count was equal to one. Downstream analysis was performed using R v3.2.2 (ref. ⁵⁰) and Python v3.6.5 (ref. ⁵¹). Graphics were generated using the R packages ggplot2 (ref. ⁵²), RColorBrewer⁵³ and ggsqlogo⁵⁴.

Calculation of enrichment ratios in DMS. The enrichment ratios of a given variant were calculated according to previous methods⁵⁵. The clonal frequencies of variants enriched for antigen specificity by FACS, $f_{i,Ag+}$, were divided by the clonal frequencies of the variants present in the original library, $f_{i,Ab+}$, according to equation (1):

$$\text{Enrichment ratio} = \frac{f_{i,Ag+}}{f_{i,Ab+}} \quad (1)$$

A minimum value of -2 was designated to variants with $\log[\text{enrichment ratio}]$ values that were less than or equal -2 and variants that were absent from the dataset were disregarded in the calculation. A clone was defined based on the exact amino-acid sequence of the CDRH3.

Codon selection for rational library design. Codon selection for rational library design was based off equation (2), where $Y_{n,deg}$ represents the amino-acid frequency for a given degenerate codon scheme, $Y_{n,target}$ is the target amino-acid frequency and n is the number of amino acids, 20. Residues identified in DMS analysis as having a positive enrichment (enrichment ratio > 1 , or $\log[\text{enrichment ratio}] > 0$) were normalized to their enrichment ratios, converted to theoretical frequencies and taken as the target amino-acid frequencies. Degenerate codon schemes that most closely reflected these frequencies were then selected, as calculated by the mean squared error between the degenerate codon and the target frequencies:

$$\text{Optimal codon} = \arg_{\min} \left(\frac{1}{n} \sum_{i=1}^n (Y_{n,deg} - Y_{n,target})^2 \right) \quad (2)$$

In certain instances, if the selected degenerate codon did not represent desirable amino-acid frequencies or contained undesirable amino acids, a mixture

of degenerate codons were selected and pooled to achieve better coverage of the functional sequence space. Single-stranded oligonucleotides containing the selected degenerate codons were then ordered and synthesized by Integrated DNA Technologies, which served as the HDM donor template.

Machine and deep-learning model construction. Machine-learning models were built in Python v3.6.5. One- and two-body logistic regressions, linear and Gaussian-based support vector classifiers, k -nearest neighbours and random forests were implemented using Scikit-learn v0.22.1 (ref. ⁵⁶). Artificial neural networks, LSTM-RNNs and CNNs were built using the Keras v2.1.6 Sequential model⁵⁷ as a wrapper for TensorFlow⁵⁸ v1.8.0. The model architecture and hyperparameters are provided in Supplementary Table 3. The CNN architecture and hyperparameters were optimized by performing a randomized search of different variables, including nodes per layer, activation function(s), optimizer, loss function, dropout rate, batch size, number of epochs, number of filters, kernel size, stride length and pool size. The accuracy of the model across different parameters was assessed by implementing a tenfold cross validation of the dataset.

Machine and deep-learning model training and testing. Datasets for antibody-expressing, non-binding and binding sequences (sequencing statistics in Supplementary Tables 1 and 2) were aggregated to form a single binding/non-binding dataset where antibody-expressing sequences were classified as non-binders, unless they were also identified among the binding sequences. Sequences from one round of antigen enrichment were excluded from the training dataset. The complete aggregated dataset was then randomly arranged and appropriate class-labelled sequences were removed to achieve the desired classification fraction of binders to non-binders (50/50, 20/80, 10/90 and non-adjusted). The class-adjusted dataset was further split into a training set (70%) and two testing sets (15% each). Additional models were trained after increasing the number of non-binding sequences present in the training data by incrementally adding 1,000 of the removed non-binding sequences to the training set. The test datasets were kept at a fraction of 50% binders and non-binders regardless of the training dataset.

Calculation of performance metrics. The model performance metrics were calculated according to equations (3)–(6), established by methods that were previously reported⁵⁹. True positives (TP), true negatives (TN), false positives (FP) and false negatives (FN) were all determined by comparing the model classification predictions of the test dataset with the known true classifications. The F1 score is the harmonic mean of precision and recall conveying the balance between these two metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + FN + FP + TN} \quad (3)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

$$\text{F1 score} = \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

The Matthew correlation coefficient (MCC) of a given model was calculated according to equation (7), established by previous methods⁶⁰:

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (7)$$

Sequence-similarity and model-attribution analysis of predicted variants.

Sequence-similarity networks of sequences predicted to be antigen-positive and antigen-negative were constructed for LDs of 1–6 using the igraph R package⁶¹ v1.2.4. The resulting networks were analysed with respect to their overall connectivity, the composition of their largest clusters and the overall degree distribution between the classes.

The Integrated Gradients technique³⁸ was used to assess the relative attribution of each feature of a given input sequence towards the final prediction score. First, a baseline was obtained by zeroing out the input vector and the path integral of the gradients from baseline to the input vector was then approximated with a step size of 100. Integrated Gradients were visualized as sequence logos. Sequence logos were created using the Python module Logomaker⁶².

In silico sequence classification and sequence parameters. All possible combinations of amino acids present in the DMS-based combinatorial mutagenesis libraries were used to calculate the total theoretical sequence space

of 7.17×10^6 . Sequence variants (7.2×10^7) were generated in silico by taking all possible combinations of the amino acids used per position in the combinatorial mutagenesis library designed from the DMS data following three rounds of enrichment for antigen-binding variants (Supplementary Figs. 2c and 3c); alanine was also selected to be included at position 103. All in silico sequences were then classified as a binder or non-binder by the trained LSTM-RNN and CNN models. Sequences were selected for further analysis if they were classified in both models with a prediction probability (P) of more than 0.75.

The Fv net charge and FvCSP were calculated as described previously³. Briefly, the net charge was determined by first solving the Henderson–Hasselbalch equation for each residue at a specified pH (here 5.5) with known amino-acid pK_a values⁶³. The sum across all residues for both the VL and VH was then calculated as the Fv net charge. The FvCSP was calculated by taking the product of the VL and VH net charges. The HI was also calculated as described previously³, according to the following equation: $HI = -(\sum n_i E_i / \sum n_j E_j)$, where E represents the Eisenberg value of an amino acid, n is the number of an amino acid, and i and j are hydrophobic and hydrophilic residues, respectively.

The protein solubility score was determined for each full-length CDRH3 sequence (15 amino acids padded with ten amino acids on both the 5' and 3' ends; 35 amino acids in total) using the CamSol method²⁹ at pH 7.0.

The binding affinities for a reference set of 26 HLA alleles³² were determined for each 15-mer contained within the ten-amino-acid-padded CDRH3 sequence (35 amino acids) using NetMHCIIpan 3.2 (ref. ³¹). The output provides a predicted affinity (in nM) for each 15-mer as well as the percentage rank, which reflects the affinity of the 15-mer compared with a set of random natural peptides. The percentage-rank measure is unaffected by the bias of certain molecules against stronger or weaker affinities and is used to classify peptides as weak or strong binders towards the specified MHC class II allele. The minimum percentage rank, the number of 15-mers with a percentage rank below ten (classification of a weak binder) and the average percentage rank were calculated across all 21 15-mers for a single CDRH3 sequence across all 26 HLA alleles.

Overall developability improvement of the antibody sequences was determined by first normalizing the FvCSP and CamSol scores, and the average NetMHCII percentage rank (avgNetMHCII) according to the range of values observed in the remaining sequences post filtering. The normalized CamSol protein solubility score was then weighted by a factor of two for its importance in determining developability. Finally, the mean across these three parameters was taken to produce the overall developability improvement score. As the sequences were filtered with the calculated values for trastuzumab, trastuzumab would have an overall developability improvement equal to zero:

$$\text{Overall developability} = \frac{1}{3} \left(\left(\frac{\text{FvCSP} - \min(\text{FvCSP})}{\max(\text{FvCSP}) - \min(\text{FvCSP})} \right) + 2 \times \left(\frac{\text{CamSol} - \min(\text{CamSol})}{\max(\text{CamSol}) - \min(\text{CamSol})} \right) + \left(\frac{\text{avgNetMHC} - \min(\text{avgNetMHC})}{\max(\text{avgNetMHC}) - \min(\text{avgNetMHC})} \right) \right) \quad (8)$$

Antibody expression and affinity measurements using BLI. Monoclonal populations of the individual variants were isolated by performing a single-cell sort. Following expansion, the supernatant of all of the variants was collected and filtered through a 0.20- μm filter (Sartorius, 16534-K). Affinity measurements were then performed on an Octet RED96e (FortéBio) with the following parameters. Anti-human capture sensors (FortéBio, 18-5060) were hydrated in conditioned media diluted 1/2 with kinetics buffer (FortéBio, 18-1105) for at least 10 min before conditioning through four cycles of regeneration consisting of 10 s incubation in 10 mM glycine, pH 1.52, and 10 s in kinetics buffer. Conditioned sensors were then loaded with $0 \mu\text{g ml}^{-1}$ (reference sensor), $10 \mu\text{g ml}^{-1}$ trastuzumab (reference sample) or hybridoma supernatant (approximately $20 \mu\text{g ml}^{-1}$) diluted 1/2 with kinetics buffer, followed by blocking with mouse IgG (Rockland, 010-0102) at $50 \mu\text{g ml}^{-1}$ in kinetics buffer. After blocking, the loaded sensors were equilibrated in kinetics buffer and incubated with either 5 or 25 nM HER2 protein (Sigma-Aldrich, SRP6405-50UG). Finally, the sensors were incubated kinetics buffer to allow antigen dissociation. Kinetics analysis was performed using the analysis software Data Analysis HT v11.0.0.50.

Thermal-stability measurements by fluorescence. Monoclonal antibodies of the individual variants were purified by Protein A column chromatography from the supernatant of their respective monoclonal cell line and eluted with 200 mM sodium dihydrogen phosphate, 140 mM sodium chloride, pH 2.5, and then neutralized by the addition of 25% vol/vol 1 M Tris, pH 9. The protein purity was verified by SDS–PAGE before downstream analysis. Purified antibody was loaded into an UNcle instrument (Unchained Labs) and static light scattering and fluorescence measurements were taken while exposing the antibody to a thermal ramp from 20 °C to 95 °C at a rate of $0.5^\circ\text{C min}^{-1}$. The melting temperature was identified as the inflection point of the first derivative of the barycentric mean as a function of the temperature.

Immunogenicity risk assessment by T-cell proliferation assay. Immunogenicity risk was assessed using a ProMap T-cell proliferation assay (ProImmune). Briefly,

15-mer peptides for specified variant sequences were synthesized and used for the in vitro assessment of potential antigenicity. Each 15-mer peptide was pulsed into donor antigen-presenting cells, which were then co-cultured with the CD4⁺ T cells of the donor. The proliferation of CD4⁺ T cells was then measured by flow cytometry. The assay was performed by testing the peptides against 20 healthy-donor cell samples. The donor cell samples were CD8-depleted before use to eliminate CD8⁺ responses from the analysis. Detection of proliferation of CD4⁺ T cells was performed by flow cytometry after labelling cells with carboxyfluorescein succinimidyl ester and co-staining with anti-human CD4 antibody. The percentage antigenicity was determined from the frequency of positive donor cells out of all donors in the study. The strength of response was calculated by taking the mean percentage stimulation above background across all donors. The response index values were determined by multiplying the percentage antigenicity by the strength of response. Significantly responding donors were determined from the number of donors with a percentage stimulation above background of $\geq 0.5\%$ and was two standard errors greater than the background (ANOVA $P \leq 0.05$).

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The main data supporting the results in this study are available within the paper and its Supplementary Information. The raw and analysed datasets generated during the study are too large to be publicly shared; however, they are available for research purposes from the corresponding author on reasonable request.

Code availability

Deep-learning models were built in Python v3.6.5 using the Keras v2.1.6 Sequential model as a wrapper for TensorFlow v1.8.0. The code and models used to perform the work in this study are available at the following github repository: https://github.com/dahjan/DMS_opt.

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Author contributions

D.M.M., S.F., C.R.W. and S.T.R. developed the methodology. D.M.M. and S.T.R. designed the experiments and wrote the manuscript. D.M.M., C.R.W., S.F. and J.D. analysed the sequencing data and performed deep-learning analyses. P.G. and B.E.C. designed and performed the structural modelling experiments and analysis. C.J. generated in silico libraries. D.M.M. and R.A.E. performed experiments. B.W., S.M.M. and L.B. performed the cell-line development.

Competing interests

ETH Zurich has filed for patent protection on the technology described herein, and D.M.M., S.F., C.R.W. and S.T.R. are named as co-inventors on this patent (International Filing Application PCT/IB2020/053370).

Additional information

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Software and code

Policy information about [availability of computer code](#)

Data collection

No software was used for data collection.

Data analysis

All software used to analyse the data is described in detail in Methods.

For the sequencing analysis: MiXCR v2.0.3 was used to perform data pre-processing of raw FASTQ files. Downstream analysis of deep mutational scanning libraries, combinatorial mutagenesis libraries, and in silico libraries were performed in R v3.2.2 or Python v3.6.5. Graphics were generated by using the ggplot2, RColorBrewer, ggseqlogo, igraph and Logomaker packages. For deep learning: all machine learning models were built in Python v3.6.5 using the Keras v2.1.6 Sequential model as a wrapper for TensorFlow v1.8.0. Binding-kinetics analysis was performed in the Data Analysis HT v11.0.0.50 software (ForteBio). Protein solubility was determined by CamSol v2.1, and peptide binding to MHC class-II molecules was determined by NetMHCIIpan v3.2.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No statistical methods were used to determine the sample size of the number of randomly selected sequence variants (42) to experimentally validate the precision of models used in this study. The sample size of 42 was selected based on the experimental effort required to generate monoclonal cell lines of each selected variant. The authors inherently determined a sample size of 42 is sufficient to convey the precision of the described models.
Data exclusions	No data were excluded.
Replication	All attempts at replication were successful. Methods and materials used for data generation, collection and analysis are described in detail and are reproducible.
Randomization	Sequence variants selected for assessing model-classification precision were randomly chosen from the list of potential 6.9×10^6 sequence variants. No other randomization was relevant or required for the remaining experiments performed.
Blinding	The study was not blinded. Group allocation during data collection was not relevant; rather, we aimed to predict antigen binding for unknown sequence variants.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology
☒	☐ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	AlexaFluor488-conjugated AffiniPure Goat Anti-Human IgG (H+L), Jackson ImmunoResearch, code number: 109-545-088, lot number: 134438 Human CellExp ErbB2 / HER2 / CD340, Sigma-Aldrich, product number: SRP6405-50UG, lot numbers: 4J10L73970, 2J267397; AlexaFluor647 and AlexaFluor488 Microscale protein labeling kits, ThermoFisher, catalog numbers: A30009, A30006
Validation	Anti-Human IgG: Based on immunoelectrophoresis and/or ELISA, the antibody reacts with whole molecule human IgG. It also reacts with the light chains of other human immunoglobulins. No antibody was detected against non-immunoglobulin serum proteins. The antibody has been tested by ELISA and/or solid-phase adsorbed to ensure minimal cross-reaction with bovine, horse and mouse serum proteins, but it may cross-react with immunoglobulins from other species. ErbB2: expressed in HEK 293 cells, >= 95% (SDS-PAGE), < 1 EU/ug endotoxin (LAL test)

Eukaryotic cell lines

Policy information about [cell lines](#)

Cell line source(s)	All cell lines originated from a single primary murine B cell/X63Ag8 fusion hybridoma. The original hybridoma cell line (WEN1.3) was obtained as a gift from Professor Annette Oxenius (ETH Zürich).
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Authentication	Cell lines generated were developed internally and therefore no authentication was necessary nor performed.
Mycoplasma contamination	All hybridoma cell lines used in this study were confirmed annually to be negative for mycoplasma contamination (Universal Mycoplasma Detection Kit, ATCC, 30–1012K).
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines were used.

Flow Cytometry

Plots

Confirm that:

- ☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
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- ☒ All plots are contour plots with outliers or pseudocolor plots.
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Methodology

Sample preparation	Hybridoma cells were first counted and the appropriate number of cells (typically 5×10^5) were pelleted by centrifugation at 250 G for 5 minutes. Cells were then washed with PBS, re-pelleted by centrifugation, and then resuspended with the labelling antibody and/or antigen in PBS for 30 minutes on ice, protected from light. Following incubation, cells were washed twice by centrifugation and resuspension in PBS (repeated). The labeling reagents and working concentrations are described in Supplementary Table 5.
Instrument	Flow-cytometry-based analysis was performed using the BD LSR Fortessa (BD Biosciences). Fluorescence activated cell sorting (FACS) was performed using the Sony SH800S (Sony).
Software	Analysis of flow-cytometry data was performed in FlowJo v10.
Cell population abundance	For bulk population FACS, typically a minimum of 1×10^5 cells were sorted. Post-sort purity was determined by flow-cytometry analysis following expansion of the sorted population(s). For monoclonal FACS, single cells were sorted and allowed to expand. Post-sort purity of monoclonal populations was verified by genomically characterizing the region of interest by PCR/Sanger sequencing, as well as phenotype characterization by flow-cytometry analysis.
Gating strategy	Live cells were identified and gated by examining the appropriate cell size determined by the forward scatter (area) and side scatter (area) measurements. Cell doublets were then excluded by identifying and gating single cell events determined by the forward scatter (height and width) measurements. Antibody expressing cells were then identified and gated determined by their fluorescence signal from the blue (488 nm) laser (530/30 nm band-pass filter, 505 nm longpass filter). Cells expressing antigen-specific antibodies were then identified and gated determined by their fluorescence signal from the red (633 nm) laser (670/14 nm band-pass filter). Boundaries between positive and negative populations for both antibody-expressing and antigen-specific cells were determined by including the following control samples in the analysis: 1) a monoclonal cell line negative for antibody expression, and subsequently also antigen-negative (Ab-/Ag-), 2) a monoclonal cell line positive for antibody expression, but negative for antigen-binding (Ab+/Ag-), and 3) a monoclonal antibody positive for antigen-specific antibody expression (Ab+/Ag+).

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